Comparison of SimGrid and LogGOPSim for Allreduce

(Recursive Doubling and Halving)

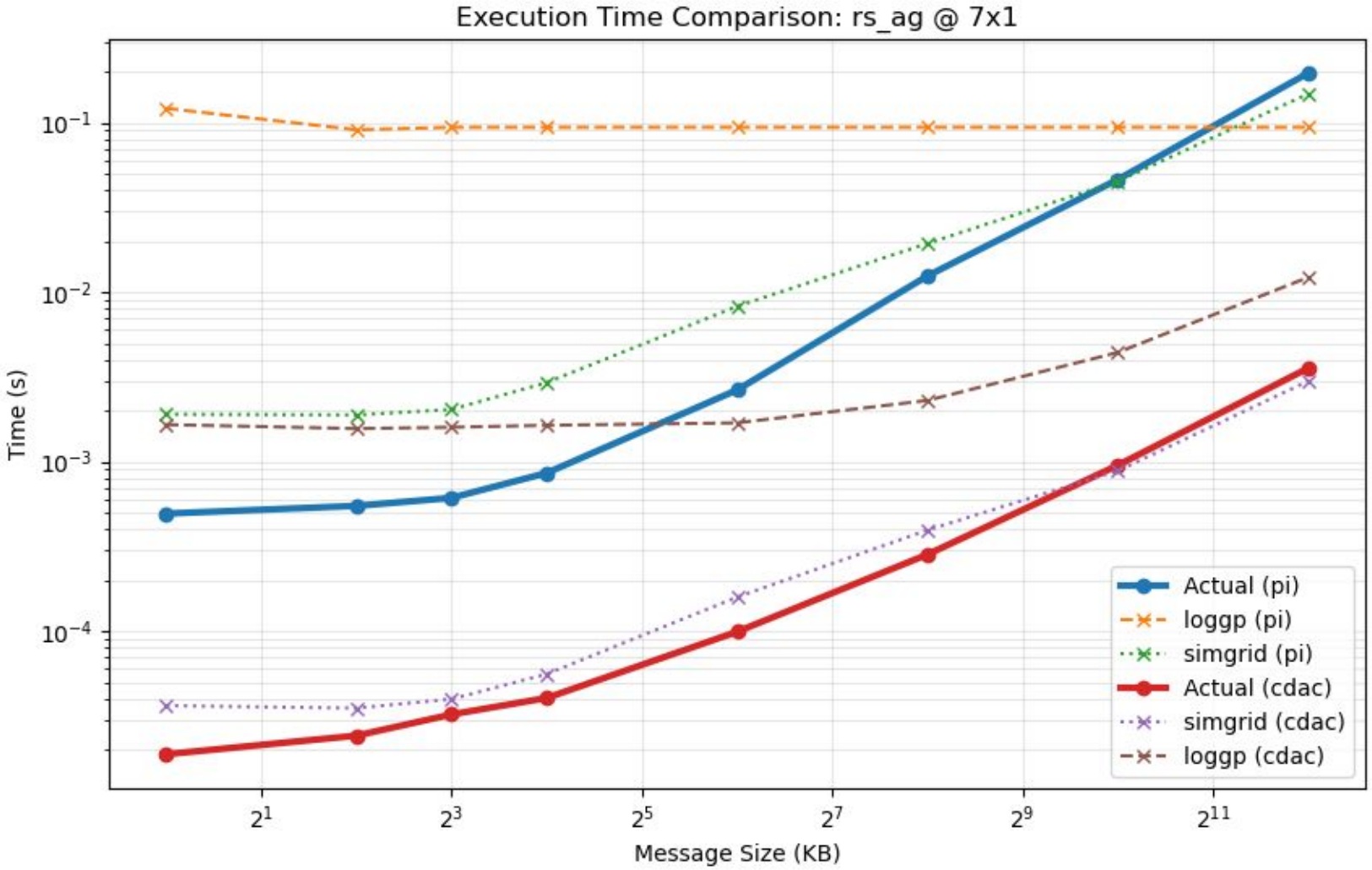
CS565: Mini Project

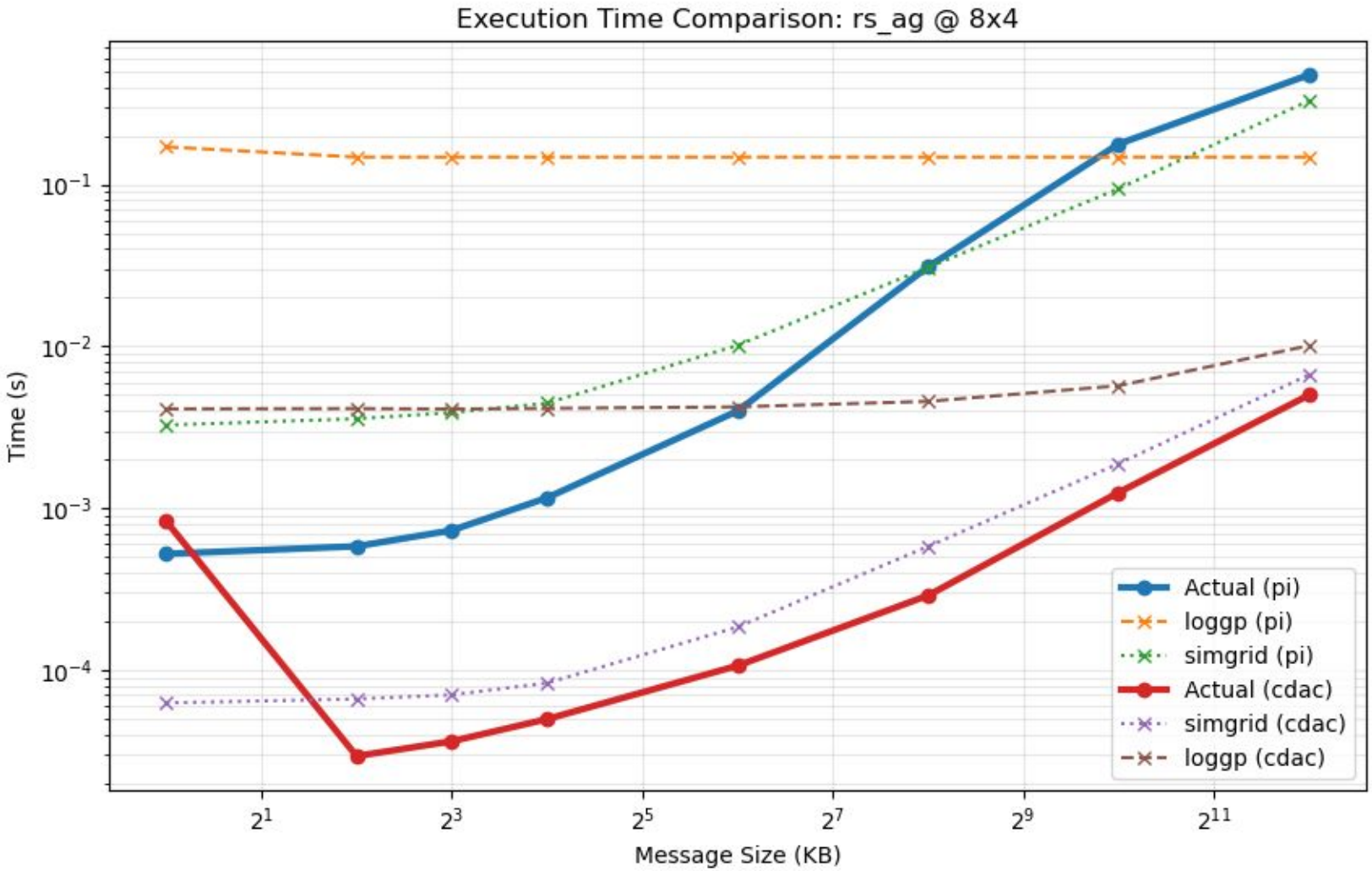
HPC Lab • CDAC & LITE-Pi Performance Analysis

Predicted vs. Observed Performance

Analyzing Simulator Accuracy across Message Size Regimes

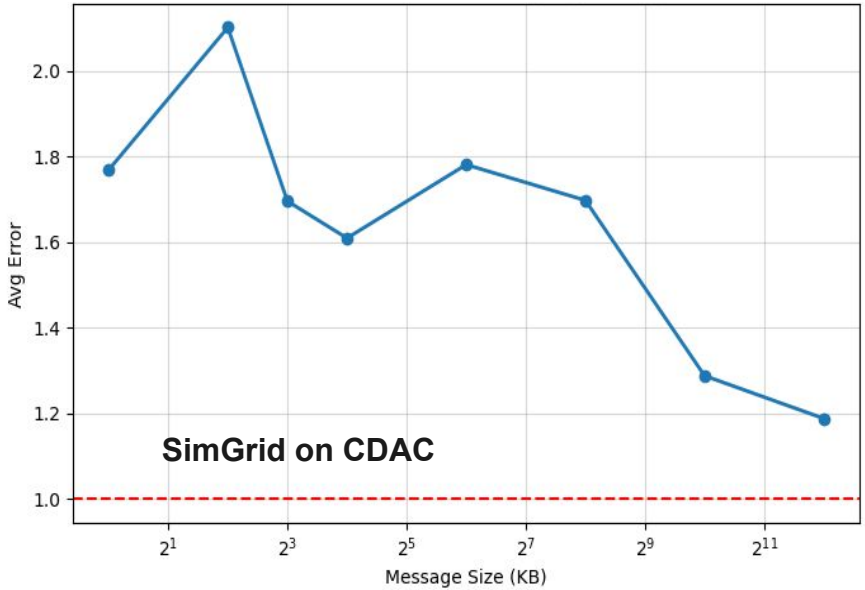


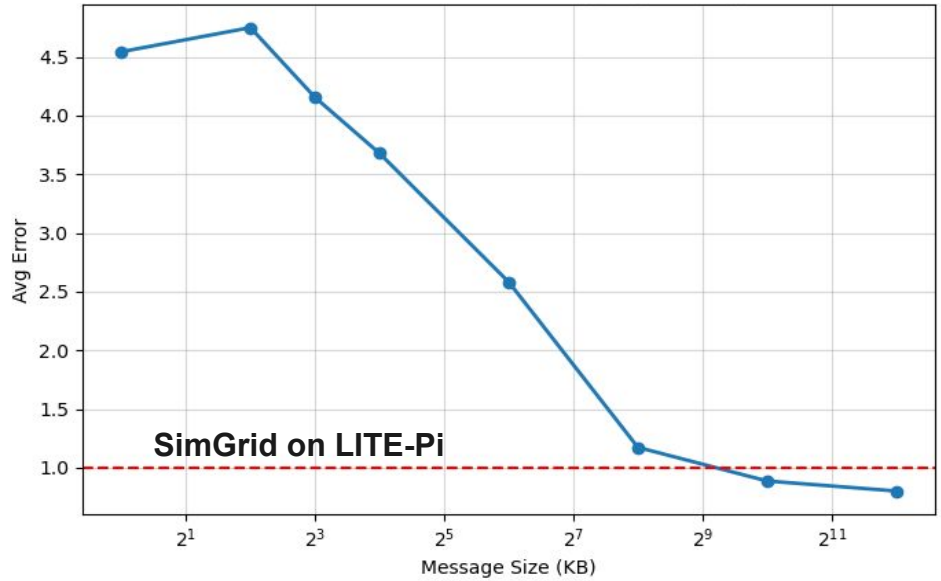

Execution Time Comparison: rs_ag @ 7x1
Time (s)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
2^1
2^3
2^5
2^7
2^9
2^{11}
Message Size (KB)
Actual (pi)
loggp (pi)
simgrid (pi)
Actual (cdac)
simgrid (cdac)
loggp (cdac)

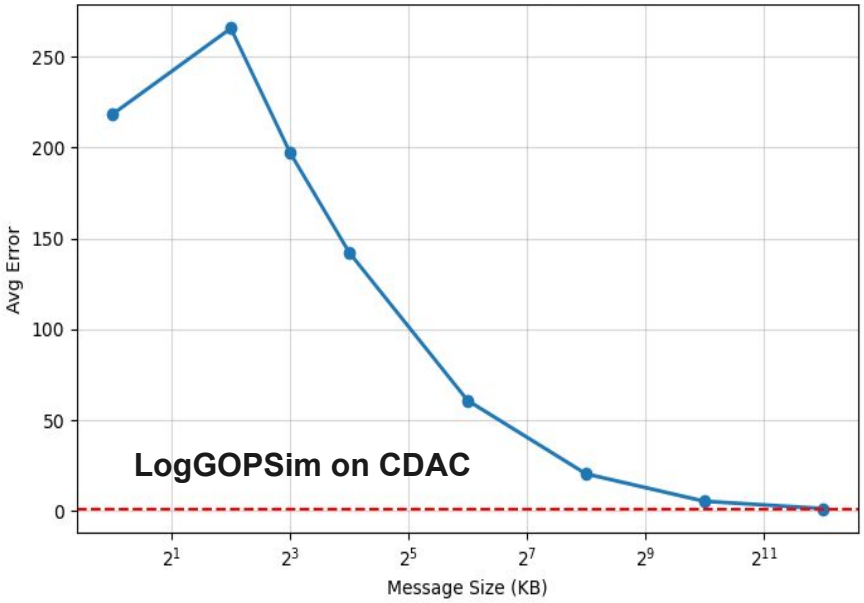

Execution Time Comparison: rs_ag @ 8x4
Time (s)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
2^{1}
2^{3}
2^{5}
2^{7}
2^{9}
2^{11}
Message Size (KB)
Actual (pi)
loggp (pi)
simgrid (pi)
Actual (cdac)
simgrid (cdac)
loggp (cdac)

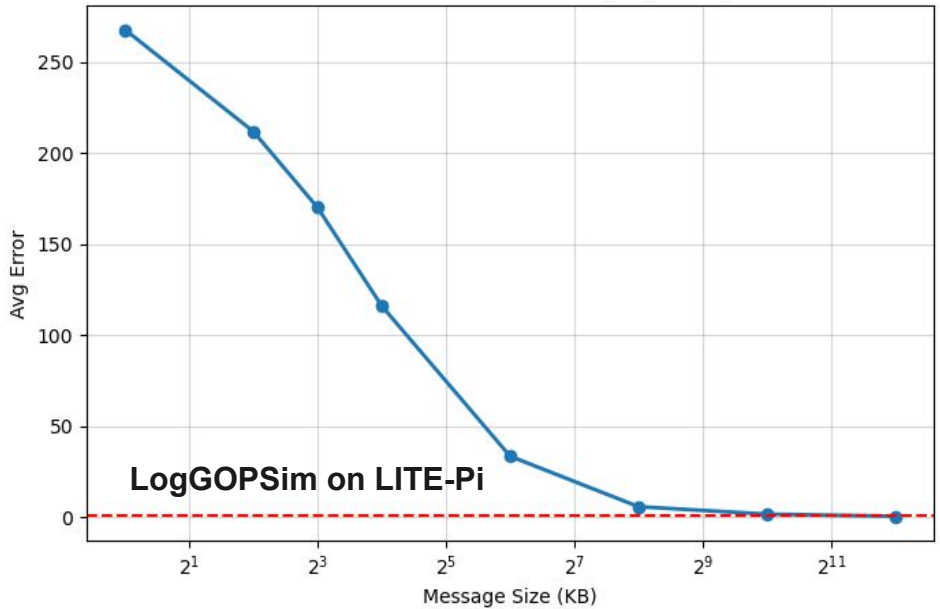
Error Sensitivity to Message Size

Average Error Variation across Message Size Regimes


Avg Error
2.0
1.8
1.6
1.4
1.2
1.0
SimGrid on CDAC
2^1
2^3
2^5
2^7
2^9
2^{11}
Message Size (KB)


Avg Error
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
SimGrid on LITE-Pi
2^1
2^3
2^5
2^7
2^9
2^{11}
Message Size (KB)

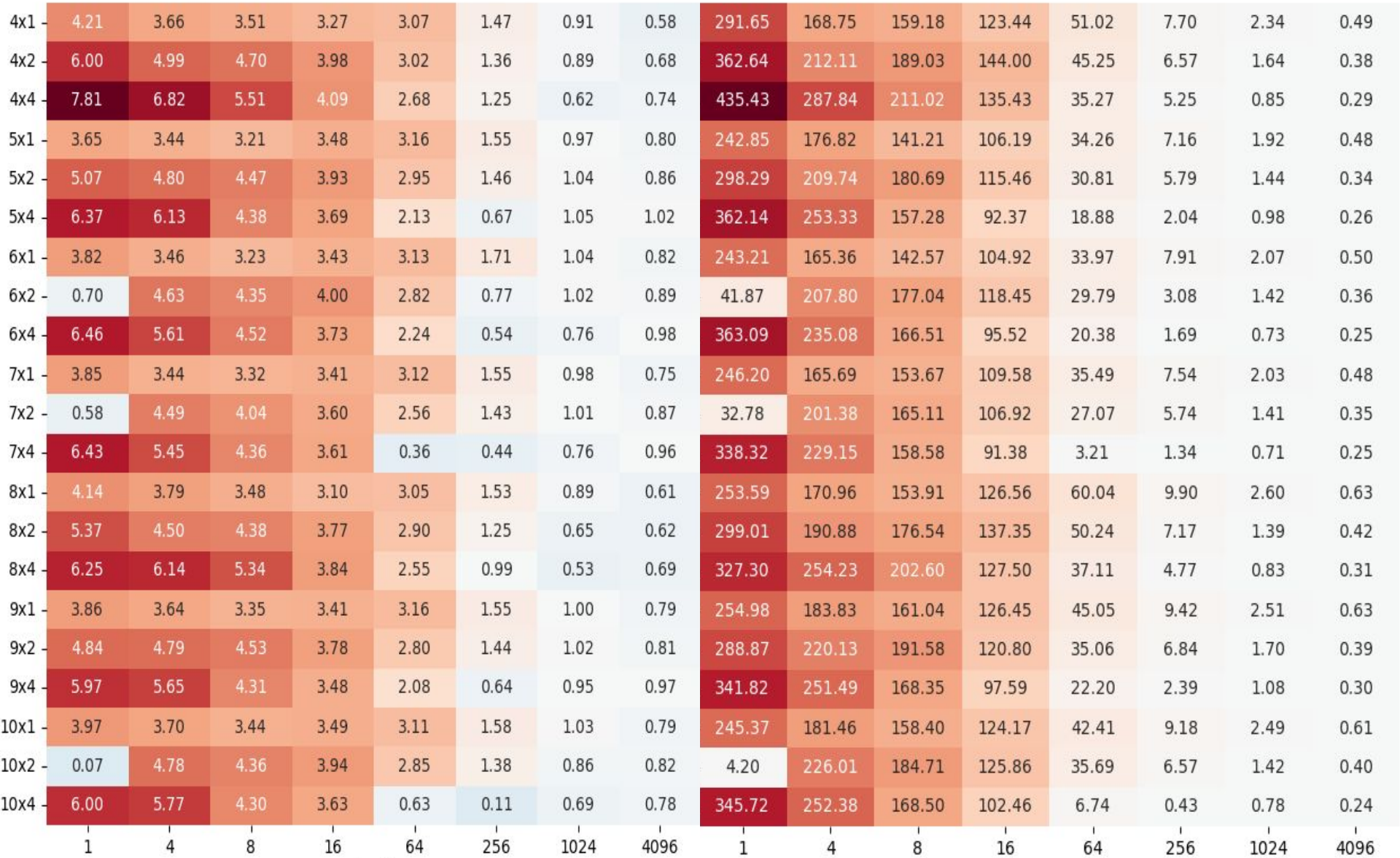

Avg Error
250
200
150
100
50
0
LogGOPSim on CDAC
2^1
2^3
2^5
2^7
2^9
2^{11}
Message Size (KB)


Avg Error
250
200
150
100
50
0
LogGOPSim on LITE-Pi
2^1
2^3
2^5
2^7
2^9
2^{11}
Message Size (KB)

Error Heatmap Analysis

SimGrid vs. LogGOPSim Performance Accuracy

Predictive Error Distribution


4x1 4.21 3.66 3.51 3.27 3.07 1.47 0.91 0.58 291.65 168.75 159.18 123.44 51.02 7.70 2.34 0.49
4x2 6.00 4.99 4.70 3.98 3.02 1.36 0.89 0.68 362.64 212.11 189.03 144.00 45.25 6.57 1.64 0.38
4x4 7.81 6.82 5.51 4.09 2.68 1.25 0.62 0.74 435.43 287.84 211.02 135.43 35.27 5.25 0.85 0.29
5x1 3.65 3.44 3.21 3.48 3.16 1.55 0.97 0.80 242.85 176.82 141.21 106.19 34.26 7.16 1.92 0.48
5x2 5.07 4.80 4.47 3.93 2.95 1.46 1.04 0.86 298.29 209.74 180.69 115.46 30.81 5.79 1.44 0.34
5x4 6.37 6.13 4.38 3.69 2.13 0.67 1.05 1.02 362.14 253.33 157.28 92.37 18.88 2.04 0.98 0.26
6x1 3.82 3.46 3.23 3.43 3.13 1.71 1.04 0.82 243.21 165.36 142.57 104.92 33.97 7.91 2.07 0.50
6x2 0.70 4.63 4.35 4.00 2.82 0.77 1.02 0.89 41.87 207.80 177.04 118.45 29.79 3.08 1.42 0.36
6x4 6.46 5.61 4.52 3.73 2.24 0.54 0.76 0.98 363.09 235.08 166.51 95.52 20.38 1.69 0.73 0.25
7x1 3.85 3.44 3.32 3.41 3.12 1.55 0.98 0.75 246.20 165.69 153.67 109.58 35.49 7.54 2.03 0.48
7x2 0.58 4.49 4.04 3.60 2.56 1.43 1.01 0.87 32.78 201.38 165.11 106.92 27.07 5.74 1.41 0.35
7x4 6.43 5.45 4.36 3.61 0.36 0.44 0.76 0.96 338.32 229.15 158.58 91.38 3.21 1.34 0.71 0.25
8x1 4.14 3.79 3.48 3.10 3.05 1.53 0.89 0.61 253.59 170.96 153.91 126.56 60.04 9.90 2.60 0.63
8x2 5.37 4.50 4.38 3.77 2.90 1.25 0.65 0.62 299.01 190.88 176.54 137.35 50.24 7.17 1.39 0.42
8x4 6.25 6.14 5.34 3.84 2.55 0.99 0.53 0.69 327.30 254.23 202.60 127.50 37.11 4.77 0.83 0.31
9x1 3.86 3.64 3.35 3.41 3.16 1.55 1.00 0.79 254.98 183.83 161.04 126.45 45.05 9.42 2.51 0.63
9x2 4.84 4.79 4.53 3.78 2.80 1.44 1.02 0.81 288.87 220.13 191.58 120.80 35.06 6.84 1.70 0.39
9x4 5.97 5.65 4.31 3.48 2.08 0.64 0.95 0.97 341.82 251.49 168.35 97.59 22.20 2.39 1.08 0.30
10x1 3.97 3.70 3.44 3.49 3.11 1.58 1.03 0.79 245.37 181.46 158.40 124.17 42.41 9.18 2.49 0.61
10x2 0.07 4.78 4.36 3.94 2.85 1.38 0.86 0.82 4.20 226.01 184.71 125.86 35.69 6.57 1.42 0.40
10x4 6.00 5.77 4.30 3.63 0.63 0.11 0.69 0.78 345.72 252.38 168.50 102.46 6.74 0.43 0.78 0.24
1 4 8 16 64 256 1024 4096 1 4 8 16 64 256 1024 4096

Error Heatmap: rs_ag | simgrid simulating cdac

	1	4	8	16	64	256	1024	4096
4x1	3.83	2.34	1.86	1.97	2.06	2.29	1.29	1.12
4x2	0.09	3.77	3.10	2.32	2.34	2.44	1.69	1.66
4x4	4.42	4.15	3.29	2.92	3.01	3.07	2.58	2.73
5x1	2.64	1.65	1.42	1.17	1.57	1.27	0.90	0.85
5x2	2.92	1.88	1.61	1.48	1.76	1.48	1.14	1.03
5x4	0.29	1.99	1.76	1.76	2.00	1.80	1.55	1.40
6x1	2.22	1.41	1.24	1.33	1.33	1.24	0.91	0.85
6x2	2.78	1.93	1.54	1.12	1.87	1.50	1.14	1.06
6x4	2.46	1.87	1.73	1.72	2.08	1.79	1.53	1.40
7x1	1.94	1.46	1.23	1.44	1.60	1.33	0.94	0.85
7x2	2.47	2.07	1.70	1.51	1.87	1.37	1.11	1.15
7x4	0.15	2.04	1.89	1.82	2.06	1.78	1.42	1.41
8x1	2.28	1.71	1.23	0.98	0.75	1.54	0.99	0.92
8x2	0.12	2.47	1.37	1.59	1.44	1.68	1.20	1.01
8x4	0.08	2.20	1.89	1.63	1.72	2.00	1.53	1.30
9x1	2.54	1.87	1.01	1.33	1.46	1.32	0.96	0.86
9x2	0.15	1.63	1.66	1.61	1.64	1.41	1.15	1.03
9x4	2.25	1.99	1.86	1.85	2.13	1.85	1.49	1.48
10x1	2.02	1.87	1.48	0.88	1.36	1.32	0.97	0.87
10x2	1.58	1.63	1.16	1.52	1.69	1.45	1.17	1.05
10x4	0.18	2.15	1.77	2.16	2.04	2.05	1.57	1.40

Error Heatmap: rs_ag | loggp simulating cdac

	1	4	8	16	64	256	1024	4096
4x1	283.88	159.04	122.68	105.42	48.67	20.32	7.92	5.18
4x2	7.41	311.33	244.06	81.88	38.47	13.98	5.52	3.62
4x4	355.97	310.94	183.79	175.66	54.01	23.95	6.67	3.59
5x1	131.79	78.73	52.45	33.10	16.86	7.57	4.59	3.61
5x2	173.34	110.19	43.27	31.05	13.57	5.90	3.86	2.19
5x4	18.14	88.65	88.12	60.42	26.06	8.77	3.44	1.80
6x1	93.70	62.77	53.93	38.15	14.23	7.18	4.57	3.67
6x2	91.59	59.83	53.92	22.80	25.20	6.01	3.04	2.24
6x4	154.13	111.67	86.30	61.40	25.61	9.11	3.29	1.88
7x1	88.13	64.87	49.48	40.63	17.09	8.11	4.63	3.43
7x2	81.61	68.23	50.76	33.97	26.31	5.58	2.94	2.43
7x4	9.39	122.28	91.73	64.89	25.93	8.63	3.04	1.83
8x1	114.03	77.67	53.77	38.47	14.47	11.83	4.94	3.66
8x2	8.03	83.12	81.92	45.46	37.26	14.87	3.49	2.58
8x4	4.94	140.49	112.99	83.26	39.77	15.84	4.57	2.02
9x1	82.24	58.25	31.64	32.04	14.39	7.15	4.40	3.50
9x2	4.07	41.79	38.77	69.67	21.67	5.48	3.78	2.30
9x4	130.14	102.16	118.77	60.49	26.34	9.38	4.10	1.84
10x1	70.62	68.02	49.52	21.53	14.10	7.66	4.36	3.46
10x2	40.60	42.03	27.07	27.86	29.35	11.20	4.40	2.22
10x4	9.12	107.82	81.41	60.60	25.72	12.14	3.40	1.63

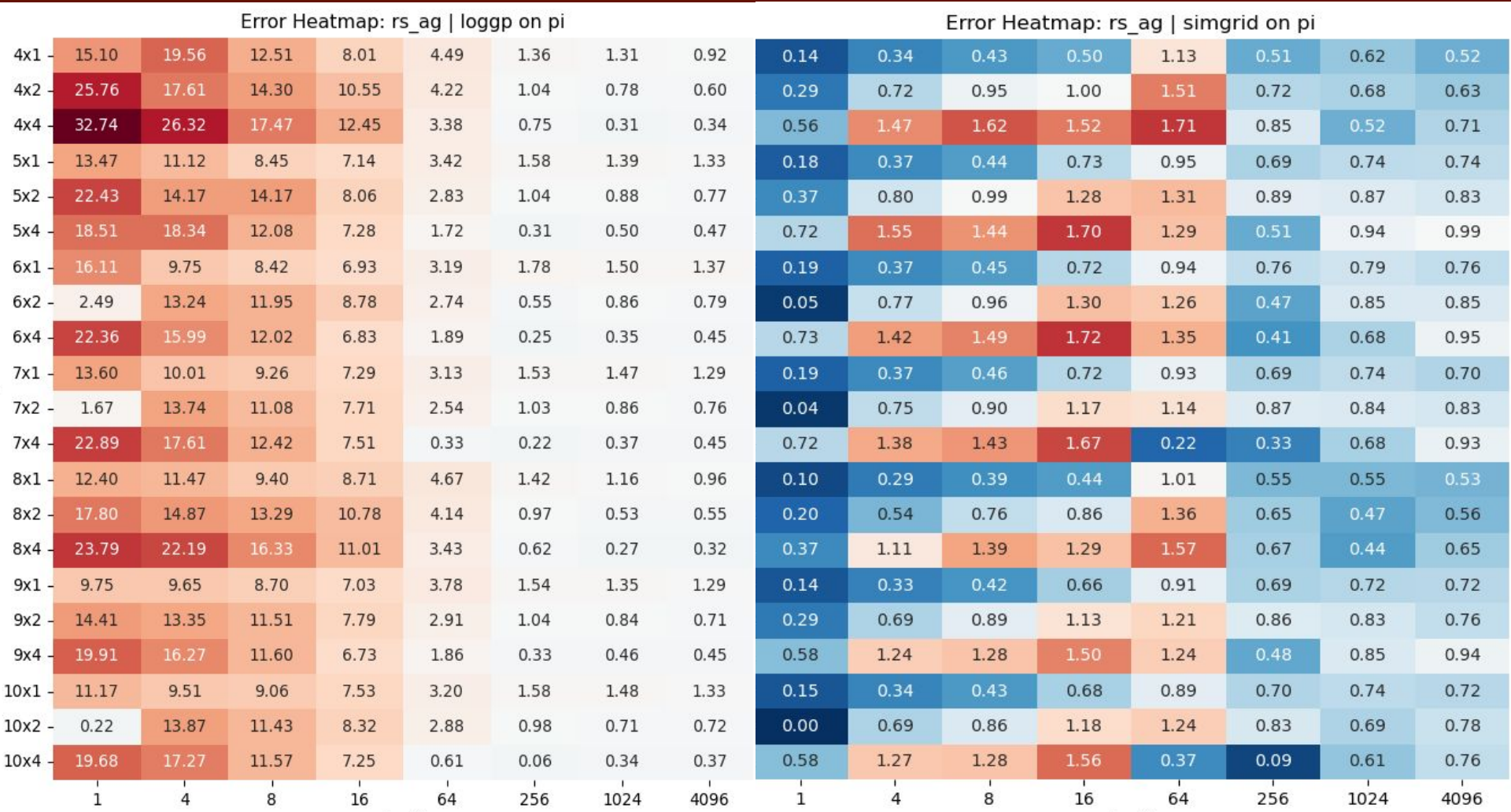
Heatmap Analysis

- Clear distinction between Eager and Rendezvous regime
- Latency (L) plays a more important role for smaller messages
- Netgauge measures L as RTT/2

.

- But MPI_Send behaves non-blockingly under eager protocol
- **Test:** Setting latency to 0 for both simulators

Heatmap Analysis



Persistent Error

- LogGOPSim still shows positive error
- Even after setting all parameters (L,o,g,G) to 0, predicted time is still far larger than actual time
- Goal files show a large calc block at the end of each rank

```
l30 requires l25
l30 requires l26
l31: calc 1537000
l31 requires l28
l31 requires l29
```

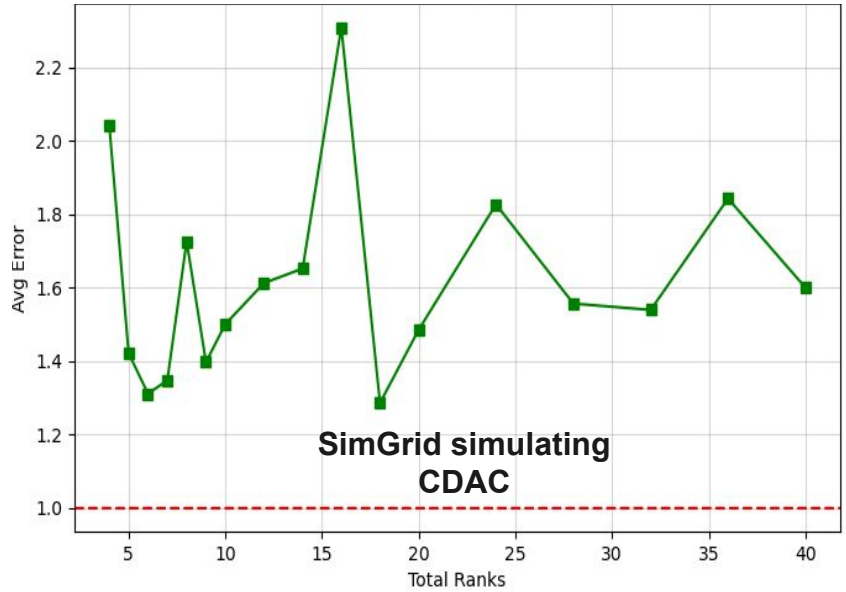
Persistent Error

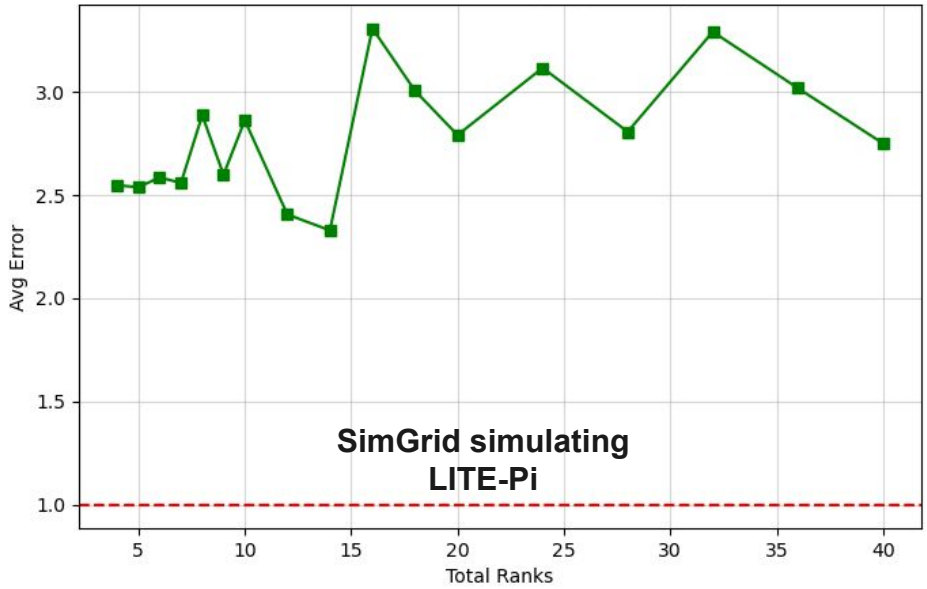
- The last calc block corresponds to cleanup (free and MPI_Finalize)
- Liballprof traces the entire program, including setup and cleanup
- Actual runtime (measured using MPI_Wtime) did not include this
- SimGrid automatically skips these functions, so it doesn’t show the same behaviour as LogGOPSim

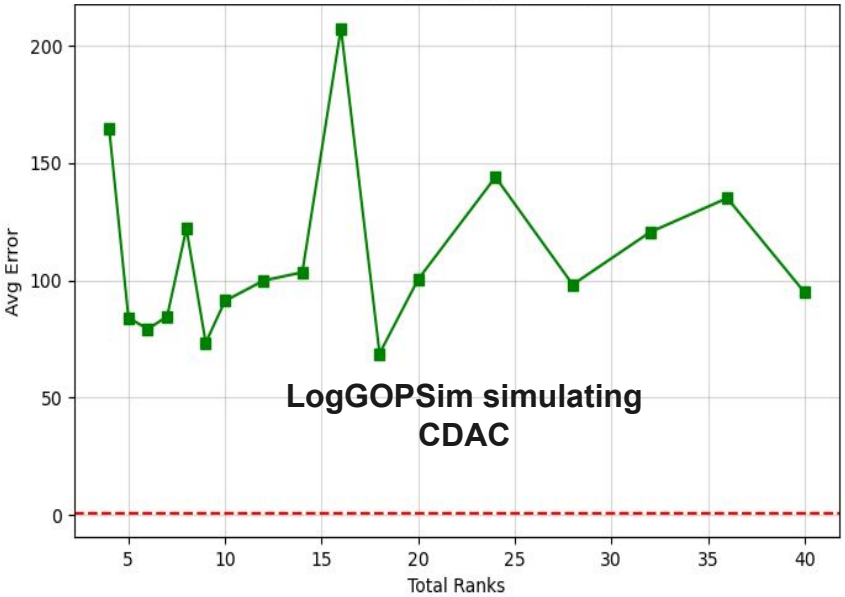
Error and Scalability

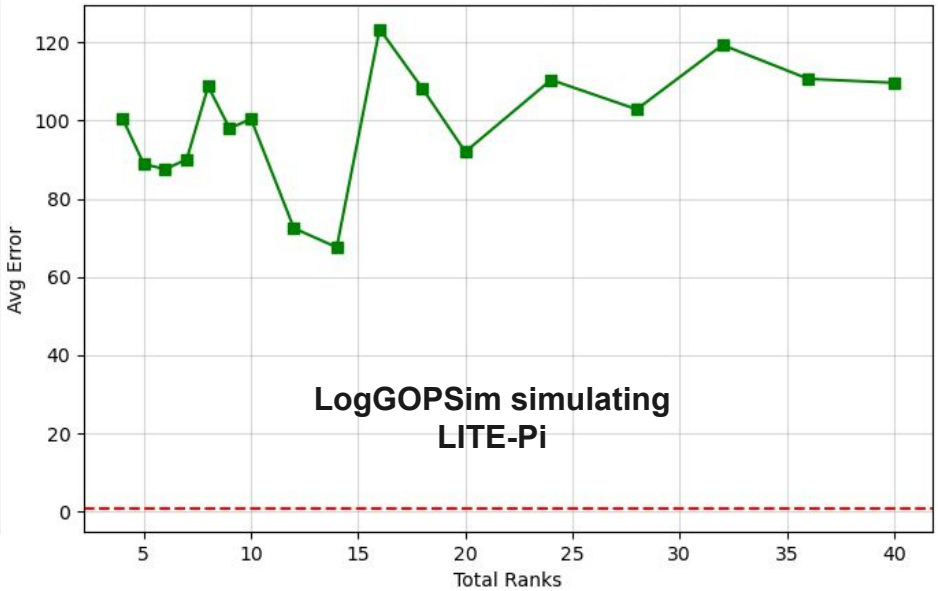
Observing the Error Ratio with increasing number of Ranks




Avg Error
Total Ranks
SimGrid simulating
CDAC


Avg Error
Total Ranks
SimGrid simulating
LITE-Pi


Avg Error
Total Ranks
LogGOPSim simulating
CDAC

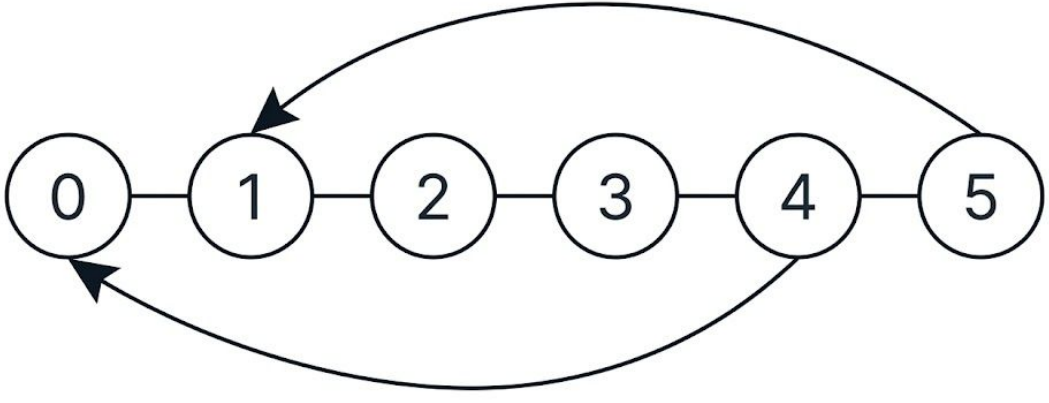

Avg Error
Total Ranks
LogGOPSim simulating
LITE-Pi

Scalability Analysis

- All graphs show similar peaks and valleys.
- For a fixed cluster, graph shapes are near identical.
- This shows that code logic dictates the shape with cluster parameters having a secondary influence.

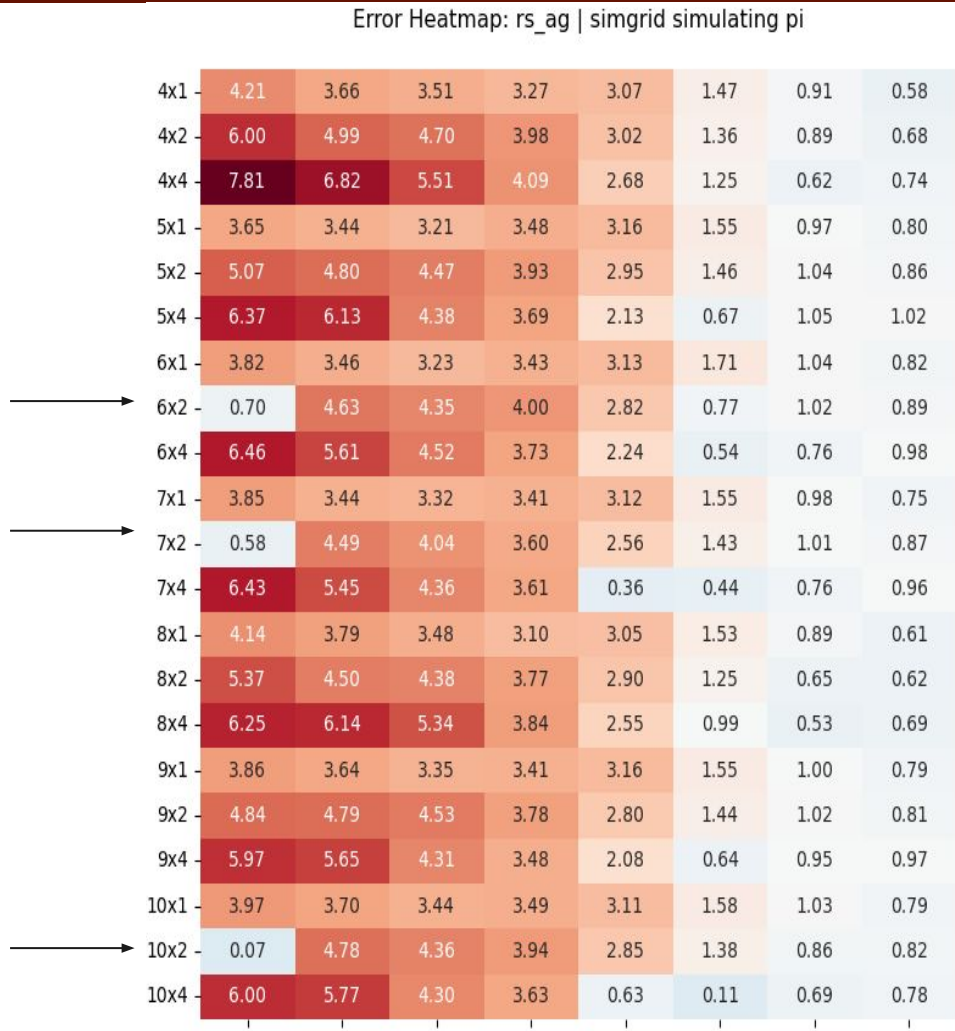
Scalability Analysis

- Eager regime dominates the average error.
- Non power of 2 cases require extra communication, which takes more time than expected.
- Because actual time spikes, the error drops for non power of 2 cases.



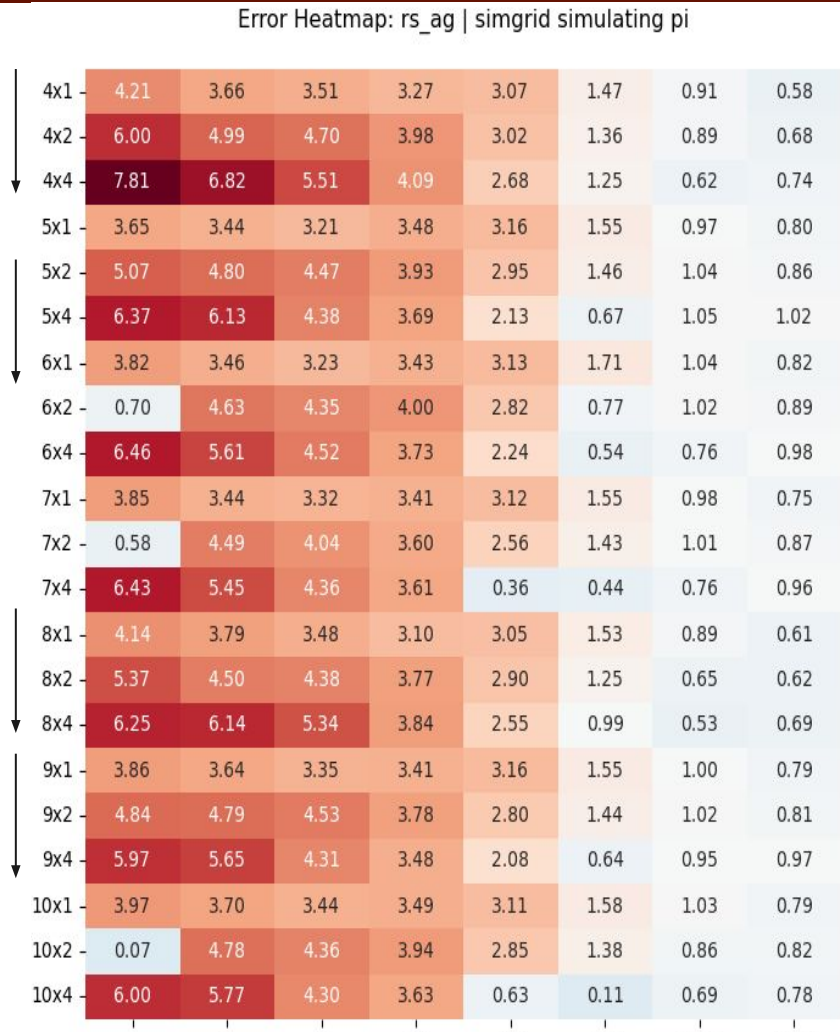
Scalability Analysis

- Best illustrated by these cells
- Overlap of:
  - Extra messages
  - Receiver Congestion
  - Limited Parallelism



Scalability Analysis

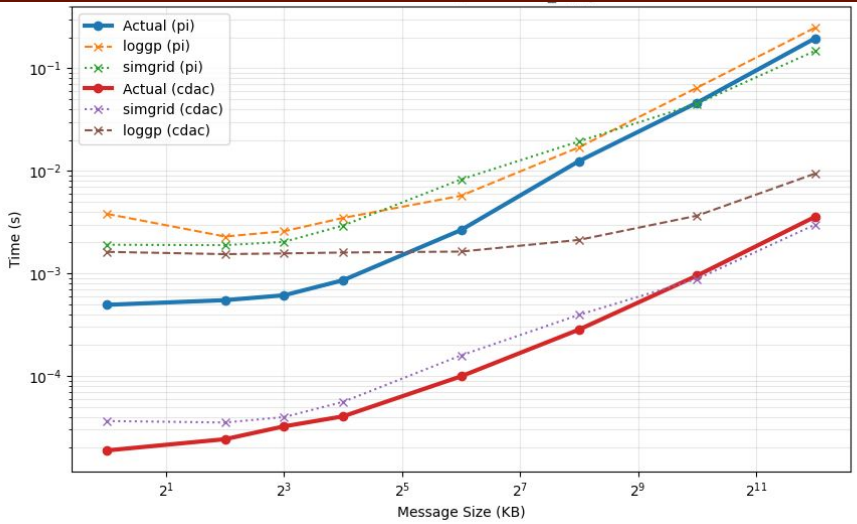
- Other cells show increasing error with increasing PPN
- Best explanation for this is simulator inaccuracy for intra-node communication
- Simulators assume intra-node takes just as much time as inter-node.

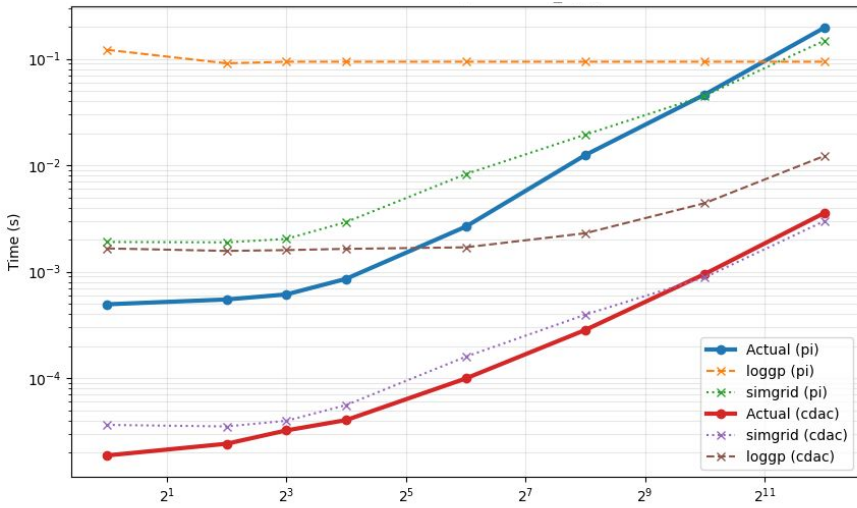


Iterative Parameters

- Netgauge provides parameter values for each msg size
- Originally, reported values were average values.

  .
- Using iterative parameters resulted in a better prediction by LogGOPSim (for pi cluster)





Thank You!